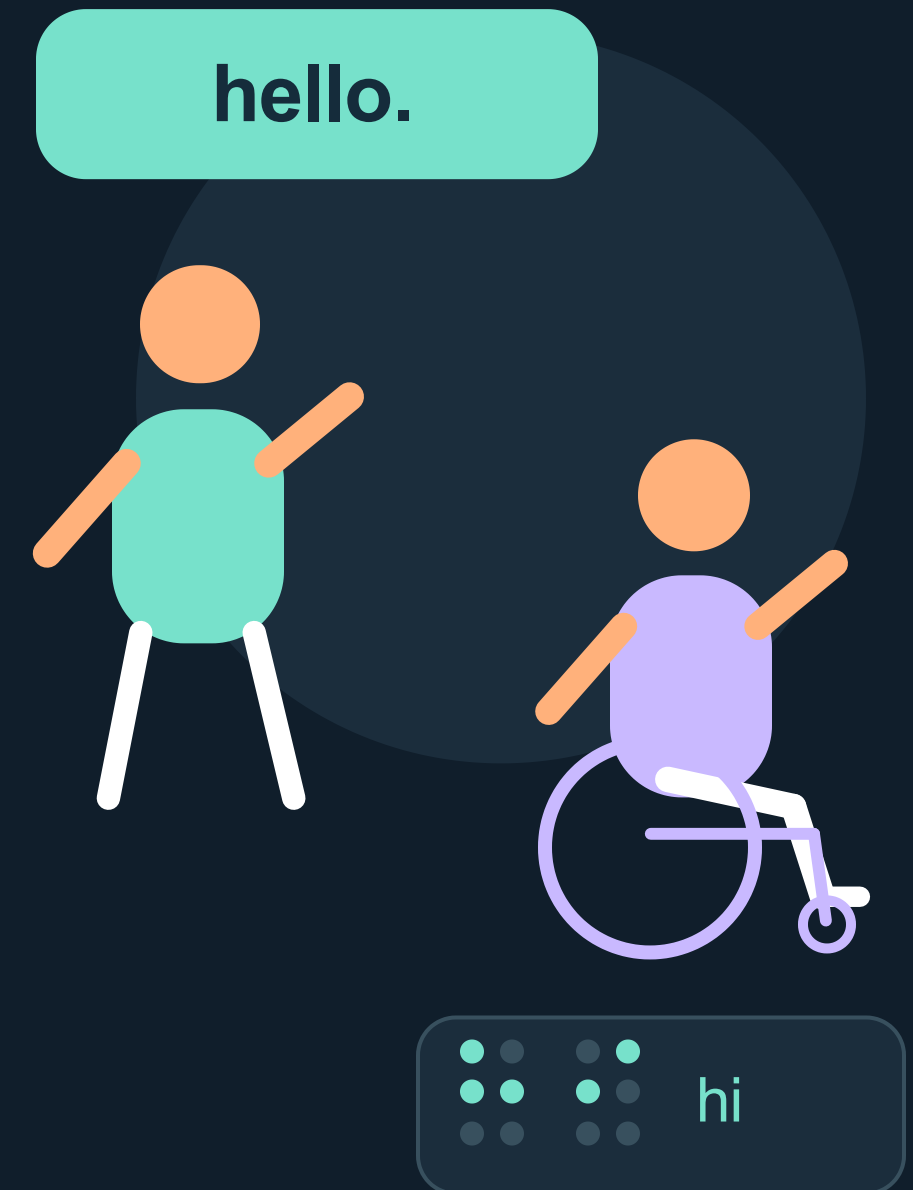


PROJECT ABILITY

A thought. A feeling. A simple “hello.”

Everyone deserves a way
to share it.

Team: [names / IDs] · [department / institution]



THE PROBLEM → OUR IDEA

Three interfaces. One shared conversation.

01 / SIGN

Glove + avatar

Gesture input for Deaf users.
Signing-avatar output planned.

02 / TOUCH

Braille + buttons

Tactile reading and character
entry for blind users.

03 / INTENT

Brain + eyes

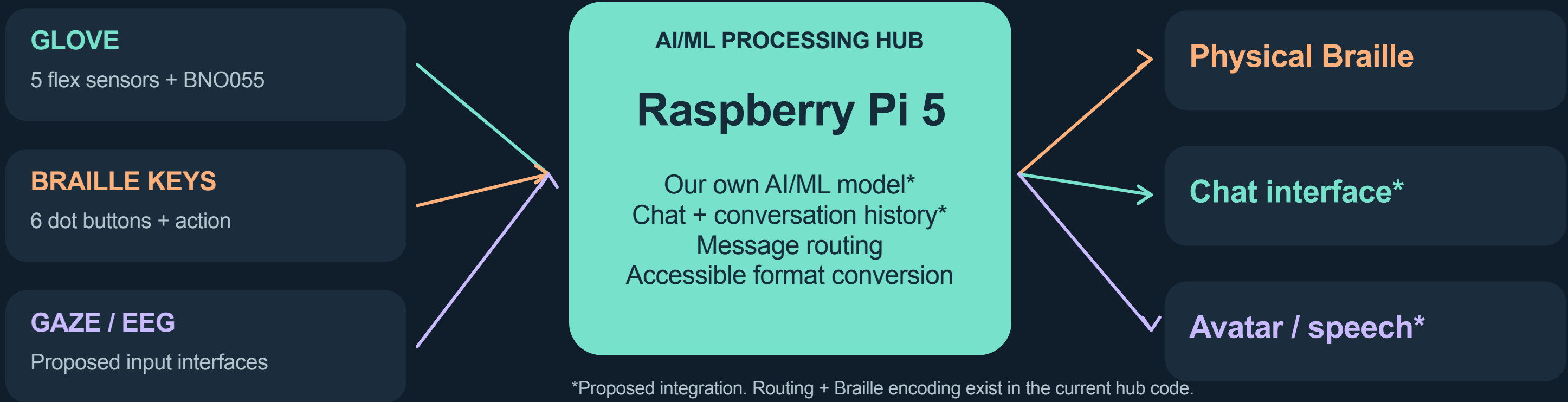
Proposed mobility and messaging
for users with motor limitations.

The contribution: connect affordable interfaces through a common message format.

OUR PLATFORM VISION

Our own AI/ML processing hub.

Raspberry Pi 5 + custom recognition model + an accessible chat interface



FEATURE 01 / GESTURE INPUT

A hand movement becomes a message.



5 flex sensors + BNO055 IMU

SENSE

Finger bend + palm orientation

INTERPRET

ESP32-S3 calibration + gesture rules

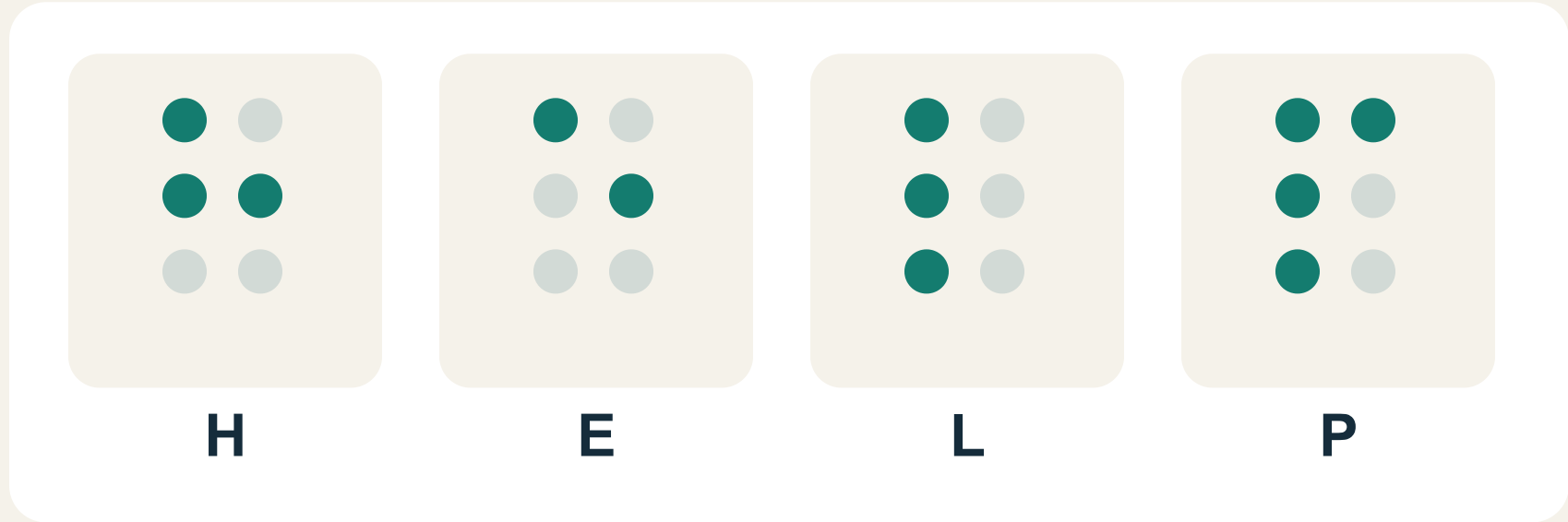
SEND

Message → MQTT → Raspberry Pi

NEXT: our own trained fusion model + a signing-avatar receiver

FEATURES 02 + 03 / TACTILE OUTPUT + BUTTON INPUT

Read with touch. Reply with buttons.



4 cells × 2 servos = 8 MG90S actuators

Cam mechanism • physical dots • replaceable parts

6 DOT KEYS + 1 ACTION



Tap dots → commit character
Double-click → space
Triple-click → local display

Hub reply / avatar link: next integration

WHERE IT MATTERS

Small moments. Real independence.

CLASSROOM

“Your turn.”

A gesture becomes tactile text for a blind classmate.

HOME

“I’m here.”

Family members exchange messages in accessible formats.

CARE / PROPOSED

“Water, please.”

A gaze keyboard composes a request without hand typing.

Designed for Deaf, blind, speech-impaired and motor-impaired users, plus their communication partners.

THE MOST IMPORTANT PART

Now, follow one message.

01

Move a hand

02

Watch the Pi route it

03

Feel the Braille

LIVE HARDWARE DEMONSTRATION →

LOW-COST DESIGN / ALL VALUES IN BDT

Built for a local budget.

GLOVE

₹4,500

Estimated range: ₹4,000–5,000

4-CELL BRAILLE

₹3,500

Target: servos + keys + mechanics

VOICE MODULE

₹1,500

Target: mic + speaker + ESP32

GAZE INTERFACE

₹3,500

Target: camera; screen + Pi shared

SHARED PI HUB

₹15,000

Target: lower-memory / reused Pi 5

WHEELCHAIR ADD-ON

₹12,000*

Control target; EEG + chair excluded

Communication target: ₹28,000

Core: ₹23,000 · With voice/gaze + 10% reserve: ₹30,800

Shared hub: ₹3,000 / user

5 users · *Mobility budget is separate

SELECTED PREMIUM REFERENCES / BDT

Premium prices. A lower-cost direction.

HAND INPUT

Rokoko Smartgloves II

≈ ₳2.45 lakh

Ability: ₳4,000–5,000

Pro motion-capture pair vs our single gesture-glove prototype.

TACTILE OUTPUT

HumanWare Brailiant BI 40X

≈ ₳4.62 lakh

Ability target: ₳3,500

40-cell commercial reader vs our 4-cell, six-dot prototype.

Lower entry cost through focused functions, shared computing and local assembly.

Price context, not equal performance. Module costs exclude hub share. USD 1 ≈ ₳123; tax/shipping excluded.
Official listings checked 30 Sep 2026. Lower-price reference: Orbit Reader 20 ≈ ₳98,300 (20 cells).

PRESENTATION-ONLY CONCEPT

PROPOSED / NOT BUILT

The next module: mobility + a voice.

PARKINSON'S / MOTOR ACCESS



Concept illustration

DRIVE MODE

EEG intent or eye selection → Pi → local safety controller → wheelchair motors

COMMUNICATION MODE / CHAIR STOPPED

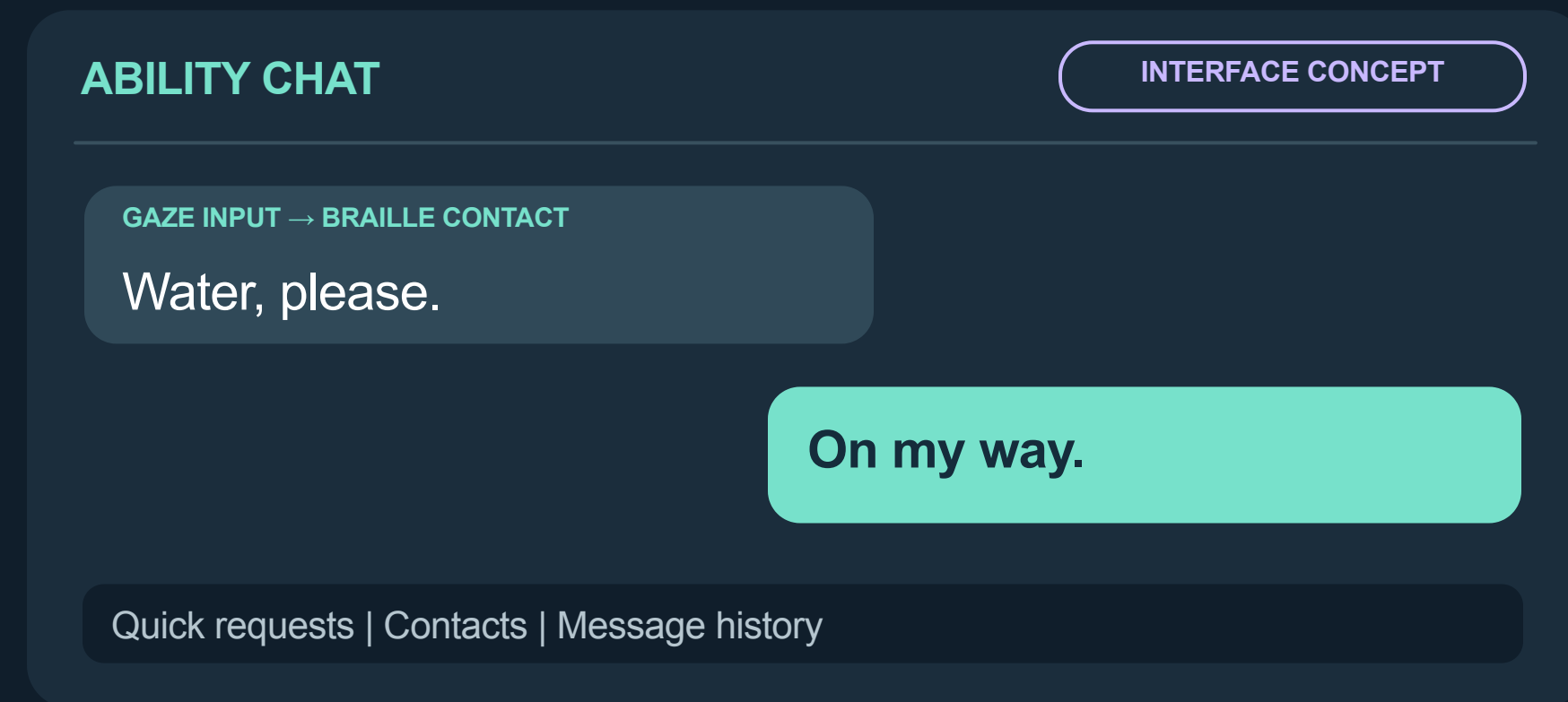
Gaze keyboard → text → Pi hub → Braille, screen, and planned avatar / speech

E-stop • obstacle stop • signal timeout • speed limit

OUR AI/ML HUB / INTERFACE CONCEPT

One hub. One accessible chat.

Recognize intent. Choose a person. Deliver in their preferred format.



OUR OWN AI/ML MODEL

Custom recognition + user calibration

ACCESSIBLE CONVERSATIONS

Text, tactile and planned avatar / speech

A CONNECTED WORKSPACE

Contacts, device status and quick requests

Next: train the model, integrate chat, and validate recognition + delivery with users.

PROJECT ABILITY

Different abilities. Shared possibilities.

Start with a message.
Build toward more independence.

MODULAR

AFFORDABLE

CONNECTED

Thank you.

BACKUP / COMPONENT COST TARGETS IN BDT

Core build: ₹23,000 target.

COMPONENT	QTY	BDT	COMPONENT	QTY	BDT
Raspberry Pi 5: 2GB / reused target	1	12,000	Glove + wiring / resistors	1	400
USB-C power supply	1	1,500	Power allowance	1	300
Active cooler	1	500	ESP32-S3	1	650
microSD storage	1	600	MG90S servos	8	1,600
Case	1	400	Cams / pins / housing	1	600
ESP32-S3	1	650	Tactile buttons	7	70
Flex sensors: economy target	5	1,250	Regulated 5V supply	1	400
BNO055: sourcing target	1	1,900	Cables / connectors	1	180

GLOVE ₹4,500 + BRAILLE ₹3,500 + SHARED HUB ₹15,000

Voice + gaze: ₹5,000 extra. Optional wheelchair control: ₹12,000 excluding EEG + chair. Full breakdown: budget.csv

BACKUP / JUDGE QUESTIONS

What the evidence supports

SOURCE PRESENT

Glove rules, MQTT routing, 4-cell actuation, button input

PARTIAL INTEGRATION

Braille keys display locally; avatar return route remains

PLANNED

Custom ML, chat UI, gaze, avatar / speech, EEG wheelchair

MEASURE NEXT

Gesture macro-F1, false triggers, tactile correctness, delay

NOVELTY POSITION

An affordable modular integration; no world-first claim

BACKUP / CLICKABLE REFERENCES

Sources & presentation scope

01 Rokoko Smartgloves II · ≈ ₺2.45 lakh · professional pair

02 HumanWare Brailiant BI 40X · ≈ ₺4.62 lakh · 40 cells

03 Bangladesh Bank · rounded presentation rate: USD 1 ≈ ₺123

04 Orbit Reader 20 · ≈ ₺98,300 · lower-price market reference

05 OpenBCI Cyton · ≈ ₺1.54 lakh · EEG board alone

06 Prior EEG/EOG wheelchair research · 2014

07 Parkinson's Foundation · vision and eye movement changes

Checked 30 Sep 2026 · Static code review, not a hardware trial.
Concept illustrations are original diagrams, not photographs of built hardware.